Predator and prey: analyzing wolf and moose populations utilizing sheaf theoretical modeling

Roberta Southwood[1], Michael Robinson[1]

[1] American University | Washington, DC 20016

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Research Goal

Our goal is to quantify the accuracy of the Lotka-Volterra predator prey system by testing how well the system performs during the time periods with unpredictable events: 1980-1982, 1996-1997, 2010s.

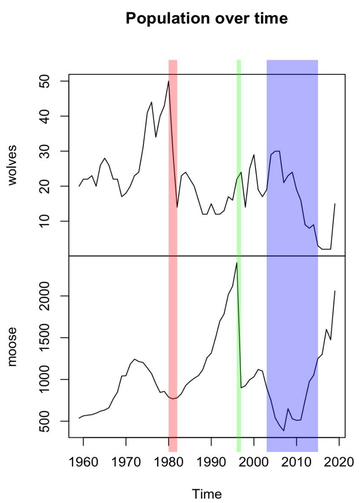
We want to answer the question:

Where does the model perform well, and where does it fall short?

Study Site

Isle Royale, Michigan

- Unique ecosystem consisting of one predator, wolves, and one main prey, moose.
- Isolated island which wolves can only get to by migrating over an ice bridge.
- Six decades of moose and wolf data (1959-2019), the longest continuous study of a predator and prey relationship in the world.[1]



Events of interest during our study years:

- 1980-1982: Wolf population crash from Global CPV2 outbreak.[2]
- 1996-1997: Moose population collapse from over foraging, ticks, and harsh winter.[1]
- 2010s: Wolf population decline from disease, inbreeding, climate change, random death of two females.[2]

Figure 1: Wolf and moose population in Isle Royale

Methods and Model

Sheaves

- A topological tool which allows mathematicians to track data locally and globally while accounting for inconsistencies and forming hypotheses about future interactions.[3]
- Can transform dynamic structural equation models (DSEM) into nonlinear models, "sheaves of dynamical systems." [3]
- Uses netlist translation of parts and nets where each arrow in the sheaf diagram represents a function.[3]

Two way model (figure 3)

- Our data was first modeled as DSEM by James T. Thorson et. al.[4]
- Parts: moose-block, moose-lag, wolves-block, and wolves-lag.
- Nets: moose, moose-pc, moose-lagvar, wolves, wolves-pc, and wolves-lagvar

Lotka–Volterra equations [5] and corresponding sheaf cells

$$\frac{dx}{dt} = \alpha x - \beta xy,$$
$$\frac{dy}{dt} = -\gamma y + \delta xy,$$

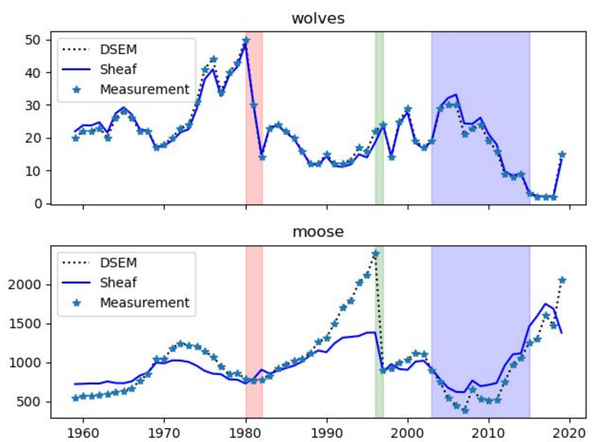
dx/dt equation

- x is population density of moose, t is time
- moose-lag and moose-lagvar come from derivative
- αx: prey have unlimited food supply and reproduce exponentially
  - moose-block to moose
- βxy: rate of predation is proportional to rate predator and prey meet
  - wolves-block to moose

dy/dt equation

- y is population density of wolves, t is time
- wolves-lag and wolves-lagvar come from derivative
- δxy: growth of predator population
  - moose-block to wolves
- γy: loss rate of the predators
  - wolves-block to wolves

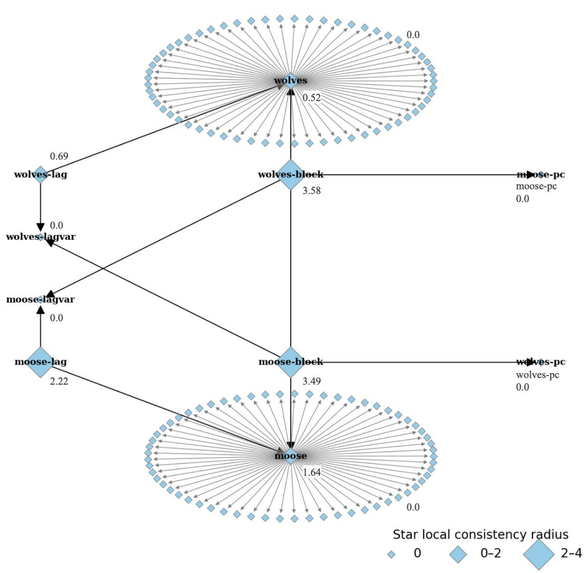
Results and Analysis



The model predicts wolves better than moose.

- Large disconnect between the DSEM and measurements compared to sheaf model around 1996-1997
  - Over foraging directly contradictions the unlimited food supply assumption of our model
- Wolves path coefficient was -0.299
- Moose path coefficient was -2.987

Figure 2: Time series of population, DSEM prediction, and sheaf prediction



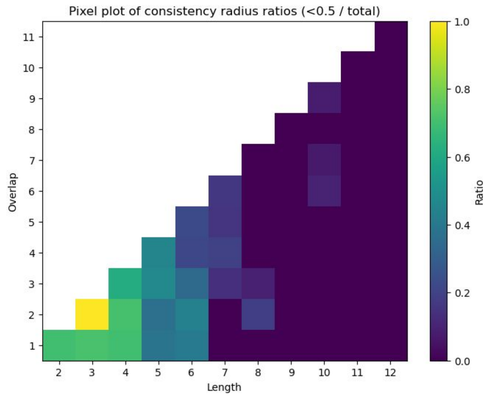
Consistency radii

- The distance that the sheaf prediction is from the recorded measurement.[3]

Figure 3

- Moose-lag and moose parts perform worse than wolves-lag and wolves parts.

Figure 3: Two way sheaf structure and resulting star consistency radii



Sliding sheaf

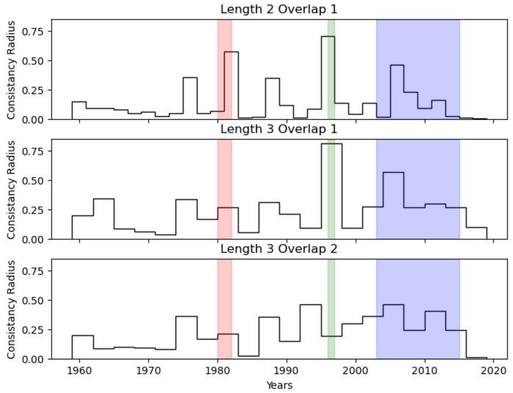
- We looked at sheaf lengths from 2 to 12 years and overlaps from 1 to 11 years.

Consistency radii

- Model starts to fail around a length of five years.

Autoregressive coefficient was also analyzed but not found to have a significant effect on our model.

Figure 6: Sliding sheaf consistency radii



The time periods with the largest consistency radii align with events the model cannot account for.

Figure 7: Consistency radii time series for lengths two and three

Conclusion and Future Work

- Moose parts (moose-lag and moose) are the weakest parts of our model.
- Model starts to fail around a length of five years.
- The time periods with the largest consistency radii align with events the model cannot account for.
- Looking forward we hope to **quantify a test statistic** to evaluate the success of our model and models to be developed.
- Continue analysis of consistency radii and parts of the sheaf model

References

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3. Robinson, M., Szulczewski, M. L.,& Thorson, T. T. (2025). Analyzing the topological structure of composite dynamical systems. https://arxiv.org/abs/2511.04603

4. Thorson J. T., Andrews A. G. III, Essington T. E., & Large S. I. (2024). Dynamic structural equation models synthesize ecosystem dynamics constrained by ecological mechanisms. *Methods in Ecology and Evolution*, 15(4), 744-755, https://doi.org/10.1111/2041-210X.14289

5. Lotka–Volterra equations. (2025). In *Wikipedia.* *https://en.wikipedia.org/wiki/Lotka%E2%80%93Volterra_equations*